

XML / DTD / XSD – Cheat Sheet (Exam Focus)

1. What is XML?

- XML = language to store & transport data (NOT programming).
- Hierarchical (tree structure).
- Used for semi-structured data (flexible vs DB tables).

Compare:

XML = data structure | HTML = display | DB = rigid tables

2. XML Syntax (VERY IMPORTANT)

- One root element
- Tags must close
- Proper nesting
- Case-sensitive

Basic structure:

```
<?xml version='1.0' encoding='UTF-8'?>
<root>...</root>
```

Elements: <name>value</name>

Attributes: <student id='1'>

Rules:

- Attribute values in quotes
- No duplicate attributes

Empty element: <tag/>

Comments: <!-- comment -->

Special characters:

< → < | > → > | & → &

CDATA: <![CDATA[text]]>

3. Well-formed vs Valid

- Well-formed = correct syntax
 - Valid = follows DTD/XSD rules
- ✓ All valid are well-formed

4. DTD (WRITE CODE)

Element declaration:

<!ELEMENT name content>

Content types:

- (#PCDATA) → text
- (A, B) → ordered children

Operators:

, = order | | = choice | ? = 0 or 1 | * = 0+ | + = 1+

Examples:

<!ELEMENT A (B, C)>

<!ELEMENT A (B | C)>

<!ELEMENT A (B*)>

Attributes:

<!ATTLIST element attr TYPE DEFAULT>

Types:

CDATA (any), ID (unique), IDREF (reference)

Defaults:

#REQUIRED, #IMPLIED, #FIXED

Example:

<!ELEMENT student (name, age)>

<!ATTLIST student id ID #REQUIRED>

DTD types:

- Internal: <!DOCTYPE root [...]>
- External: <!DOCTYPE root SYSTEM 'file.dtd'>

5. XSD (THEORY ONLY)

- XML Schema (replacement of DTD)
- Written in XML
- Supports data types (int, string...)

Key words:

schema, element, type, namespace

Example:

<xs:element name='age' type='xs:int'/>

6. XML Ecosystem

- XPath → select elements
- XQuery → query like SQL
- XSLT → transform XML

Exam Focus

- ✓ XML: fix/write structure
- ✓ DTD: write elements + attributes + operators
- ✓ XSD: theory questions only